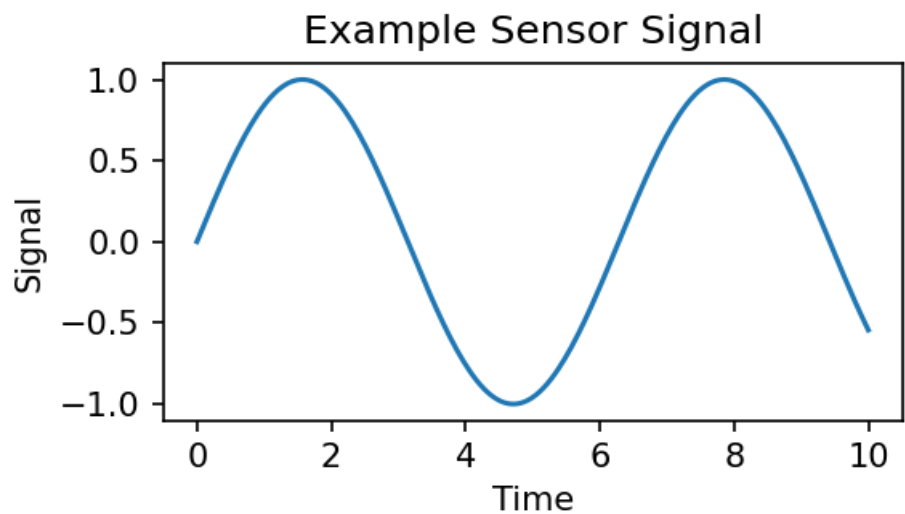


Renewable Energy: A Brief Overview

Renewable energy is produced from naturally replenishing resources such as sunlight, wind, flowing water, and geothermal heat. These resources help reduce greenhouse gas emissions and improve long-term energy security. Modern renewable technologies are becoming more affordable and efficient, making them an increasingly important part of the global electricity mix.

Illustration 1: Example Sensor Output



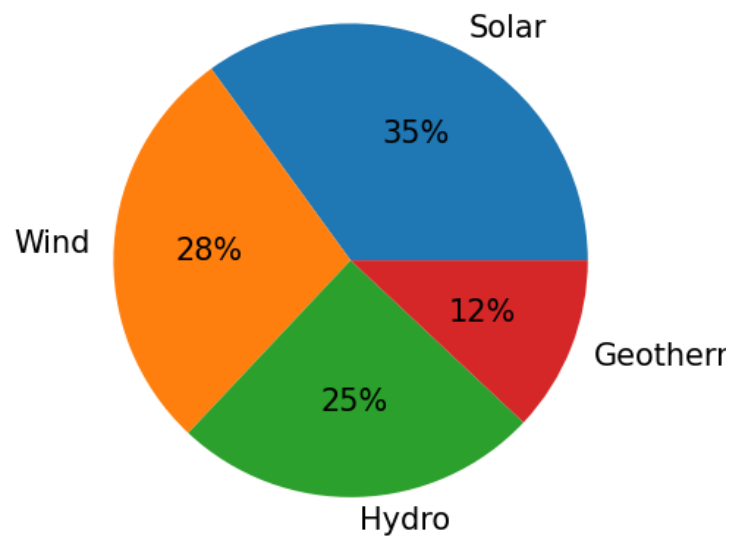
Comparison of Renewable Sources

Source	Typical Use	Advantages	Limitations
Solar	Electricity	Low maintenance	Weather dependent
Wind	Electricity	Low emissions	Variable wind speed
Hydro	Electricity	Reliable output	High infrastructure cost
Geothermal	Heating & Power	Stable supply	Location specific

Investment in renewable energy has accelerated because of falling technology costs, government incentives, and growing awareness of climate change. Energy storage systems and smart grids are improving the reliability of renewable power, allowing electricity generated during peak production periods to be stored and used later when demand rises.

Illustration 2: Sample Energy Distribution

Renewable Energy Mix



Conclusion

Renewable energy technologies continue to evolve rapidly. Combining solar, wind, hydro, and geothermal resources with energy storage and efficient transmission systems provides a pathway toward a cleaner and more sustainable energy future.